

COURTFLOW-GS: Online Multi-View Dynamic Gaussian Reconstruction with Fused MLS Tracking

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Abstract

COURTFLOW-GS reconstructs a frame-0 canonical Gaussian scene from a calibrated basketball-camera ring, then tracks dynamic motion with group-aware control points, fused rigid moving-least-squares (MLS), mask/centroid constraints, focused refinement, and periodic keyframes. The final 300-frame diagnostic produces novel-view videos and baked Gaussian checkpoints. All timing below is measured on an RTX 3080 12 GB; RTX PRO 6000 fields are intentionally reserved for replication.

1 Problem and data

The data contains 36 synchronized 4K ring cameras with fixed known calibration. Views $0, 3, \dots, 33$ train the model; the other 24 are held out. Camera poses are never optimized. The raw-capture adapter accepts `cameras/view_XXX.mp4` and `calibration/cameras.txt`, `images.txt`, producing half-resolution pinhole inputs.

2 Methodology

2.1 Canonical frame-0 reconstruction

Stage A undistorts images, detects people, prompts SAM2 masks, and associates instances through 3D occupancy carving. For each person, an occlusion-aware visual hull intersects the backprojected mask cones. A view does not carve a candidate voxel when a known foreground/static occluder explains its missing silhouette. MAST3R runs in person-specific crops; matches are triangulated with fixed cameras and rejected for poor re-projection, weak baseline, hull inconsistency, or cross-view colour inconsistency. The accepted points initialize separate person Gaussian models and a background model.

2.2 Group-aware control-point MLS

Each Gaussian is bound to its $K = 8$ nearest controls from the *same* group. Its fixed radial basis weights are

$$w_i(g) = \frac{\exp(-\|x_g - c_i\|^2/2\sigma^2)}{\sum_{j=1}^K \exp(-\|x_g - c_j\|^2/2\sigma^2)}.$$

Thus player splats cannot be pulled by other players or background controls. At each frame the optimizer estimates one shared translation per movable group plus local control offsets. A fused CUDA rigid-MLS kernel transforms Gaussian means, orientations, and degree-1 SH coefficients. The loss combines $L_1 + \text{D-SSIM}$, per-instance mask BCE, ARAP using local MLS rotations, temporal acceleration, and triangulated mask-centroid anchors. Dynamic-only deformation freezes the background controls between keyframes and composites dynamic splats over a cached background.

Architecture diagram placeholder

Suggested diagram: 12-camera images → SAM2 masks/centroid anchors → group controls → fused MLS → dynamic/background composite → focused refinement and periodic keyframe.

Figure 1: Reserved space for a final CourtFlow-GS system diagram.

2.3 Focused refinement

After every non-keyframe deformation, dynamic Gaussians are projected into every camera. Their padded, tile-aligned union crop is refined for 100 Gaussian iterations without densification. Every tenth frame receives 500 full-image iterations with bounded densification and a Gaussian-control rebind. The model state is baked forward after each frame.

2.4 Pseudocode

```
state, controls = build_canonical_scene(frame0, fixed_calibration)
for frame in sequence:
    images = decode_undistort_normalize(training_videos, frame)
    if frame % 5 == 0:
        masks = SAM2_reprompt(images, project(state.dynamic))
        anchors = triangulate(mask_centroids(masks))
        offsets = optimize(group_translation + local_control_offsets,
                           L1_DSSIM + mask_BCE + ARAP + temporal + anchors)
        state = bake(fused_rigid_MLS(state, controls, offsets))
    if frame % 10 != 0:
        state = refine(dynamic_content_crops(state, images), 100, no_densify)
    else:
        state = refine(full_images, 500, bounded_densify)
        controls = rebind(state, controls); cache_background(state)
    save_metrics_checkpoint_and_novel_view_render(state)
```

3 Results

Table 1: Quality summary. Temporal held-out samples coincide with keyframes.

Method	Full PSNR	Person PSNR	Person IoU	Measurement
Global 3DGS baseline	17.50	—	—	frame-0 held-out
Stage A canonical	19.32	18.34	0.788	frame-0 held-out
Stage B v1	18.29	15.00	0.70→0.30	keyframe held-out mean
Stage B v2 (old)	18.87	19.62	0.71→0.39	keyframe held-out mean
Focused refinement, frames 1–20	22.42	26.76	0.746	training-view mean
Final 300-frame diagnostic	20.07	22.44	—	held-out mean



Ground truth, view 13, frame 200



Earlier Stage B v2



CourtFlow-GS diagnostic

Figure 2: Same-camera comparison at held-out view 13, frame 200. Replace or add comparison rows as desired.

4 RTX 3080 runtime

Table 2: Runtime.

Component	RTX 3080 12 GB	RTX PRO 6000
MLS deformation/control optimization	1.95 s/frame	[measure on RTX PRO 6000]
Focused crop refinement	3.41 s/non-keyframe	[measure on RTX PRO 6000]
500-step full keyframe	23.16 s/keyframe	[measure on RTX PRO 6000]
Mask refresh, amortized	0.33 s/frame	[measure on RTX PRO 6000]
End-to-end diagnostic	7.61 s/frame (0.131 FPS)	[measure on RTX PRO 6000]
GPU SH colour + baked raster, 937×527	4.36 ms (229 FPS)	[measure on RTX PRO 6000]

The fused MLS/SVD path is a small fraction of tracking time; rasterization forward/backward dominates optimization. Baked-state GPU rendering is fast, while Python-to-ffmpeg transfer limits the current encoded-video exporter.

5 Limitations and next steps

Crowded late-game plays cause independent per-camera SAM2 identities to swap or disappear. In the final diagnostic, training IoU falls from 0.789 near the start to 0.001 at frame 299; held-out person PSNR falls from 23.86 to 20.59 dB. The highest-priority improvement is detector-led, multi-view identity association at keyframes. Additional work includes checkpoint consolidation or LOD for the 2.01M-Gaussian final state, unbiased inter-keyframe held-out evaluation, and direct GPU-to-NVENC output.